

```

Last login: Wed Jun 10 10:23:20 on ttys001
adityadave@Adityas-MacBook-Air-3 ~ % psql -d postgres
psql (18.4)
Type "help" for help.

postgres=# \c udf_testing
You are now connected to database "udf_testing" as user "adityadave".
udf_testing=# RESET ROLE;

ALTER TABLE orders ENABLE ROW LEVEL SECURITY;

DROP POLICY IF EXISTS orders_rls_policy ON orders;

CREATE POLICY orders_rls_policy
ON orders
USING (
    EXISTS (
        SELECT 1
        FROM customer c
        WHERE c.c_custkey = orders.o_custkey
        AND c.c_acctbal > 5000
    )
);

SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT *
FROM orders;
WARNING: ===== RAW PARSE TREE =====
DETAIL:  {RAWSTMT
:stmt
  {VARIABLESETSTMT
:kind 4
:name role
:args <>
:jumble_args false
:is_local false
:location -1
}
:stmt_location 0
:stmt_len 10
}

WARNING: ===== ANALYZED QUERY =====
DETAIL:  {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
  {VARIABLESETSTMT
:kind 4
:name role
:args <>
:jumble_args false
:is_local false
:location -1
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 10
}

RESET
WARNING: ===== RAW PARSE TREE =====
DETAIL:  {RAWSTMT
:stmt
  {ALERTABLESTMT
:relation
  {RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 12
}
:cmds (
  {ALERTABLECMD
:subtype 54
:name <>
:num 0
:newowner <>
:def <>
:behavior 0
:missing_ok false
:recurse false
}
)
:objtype 41
:missing_ok false
}
:stmt_location 0
:stmt_len 44
}

WARNING: ===== ANALYZED QUERY =====
DETAIL:  {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
  {ALERTABLESTMT
:relation
  {RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true

```

```

:relpersistence p
:alias <>
:location 12
}
:cmds (
{ALERTTABLECMD
:subtype 54
:name <>
:num 0
:newowner <>
:df <>
:behavior 0
:missing_ok false
:recurse false
}
)
:objtype 41
:missing_ok false
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 44
}

ALTER TABLE
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{DROPSMT
:objects (("orders" "orders_rls_policy"))
:removeType 28
:behavior 0
:missing_ok true
:concurrent false
}
:stmt_location 0
:stmt_len 49
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{DROPSMT
:objects (("orders" "orders_rls_policy"))
:removeType 28
:behavior 0
:missing_ok true
:concurrent false
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 49
}

DROP POLICY
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{CREATEPOLICYSTMT
:policy_name orders_rls_policy
:table
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 35
}
}
}

```

```

    }
:cmd_name all
:permissive true
:roles (
  {ROLESPEC
    :roletype 4
    :rolename <>
    :location -1
  }
)
:qual
  {SUBLINK
    :sublinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
      {SELECTSTMT
        :distinctClause <>
        :intoClause <>
        :targetList (
          {RESTARTGET
            :name <>
            :indirection <>
            :val
              {A_CONST
                :val 1
                :location 78
              }
            :location 78
          }
        )
        :fromClause (
          {RANGEVAR
            :catalogname <>
            :schemaname <>
            :relname customer
            :inh true
            :relpersistence p
            :alias
              {ALIAS
                :aliasname c
                :colnames <>
              }
            :location 93
          }
        )
        :whereClause
          {BOOLEXP
            :boolexp and
            :args (
              {A_EXPR
                :name ("=")
                :lexpr
                  {COLUMNREF
                    :fields ("c" "c_custkey")
                    :location 118
                  }
              :rexpr
                {COLUMNREF
                  :fields ("orders" "o_custkey")
                  :location 132
                }
              :rexpr_list_start 0
              :rexpr_list_end 0
              :location 130
            )
            {A_EXPR
              :name (">")
              :lexpr
                {COLUMNREF
                  :fields ("c" "c_acctbal")
                  :location 163
                }
              :rexpr
                {A_CONST
                  :val 5000
                  :location 177
                }
              :rexpr_list_start 0
              :rexpr_list_end 0
              :location 175
            )
          }
        :location 159
      }
    :groupClause <>
    :groupDistinct false
    :havingClause <>
    :windowClause <>
    :valuesLists <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :lockingClause <>
    :withClause <>
    :op 0
    :all false
    :larg <>
    :rarg <>
  }
  :location 54
}
:with_check <>
}
:stmt_location 0
:stmt_len 189
}

```

WARNING: ===== ANALYZED QUERY =====

```

DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
  {CREATEPOLICYSTMT
    :policy_name orders_rls_policy
    :table
      {RANGEVAR
        :catalogname <>
        :schemaname <>
        :relname orders
        :inh true
        :relpersistence p
        :alias <>
        :location 35
      }
    :cmd_name all
    :permissive true
    :roles (
      {ROLESPEC
        :roletype 4
        :rolename <>
        :location -1
      }
    )
    :qual
      {SUBLINK
        :sublinkType 0
        :subLinkId 0
        :testexpr <>
        :operName <>
        :subselect
          {SELECTSTMT
            :distinctClause <>
            :intoClause <>
            :targetList (

```

```

{RETARGET
:name <>
:indirection <>
:val
  {A_CONST
  :val 1
  :location 78
  }
:location 78
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname customer
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname c
:colnames <>
}
:location 93
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("c" "c_custkey")
:location 118
}
}
:rexpr
{COLUMNREF
:fields ("orders" "o_custkey")
:location 132
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 130
}
{A_EXPR
:name (">")
:lexpr
{COLUMNREF
:fields ("c" "c_acctbal")
:location 163
}
}
:rexpr
{A_CONST
:val 5000
:location 177
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 175
}
)
:location 159
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valueslists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:top 0
:all false
:larg <>
:rarg <>
}
:location 54
}
:with_check <>
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 189
}

```

CREATE POLICY
WARNING: ==== RAW PARSE TREE =====
DETAIL: {RAWSTMT

```

:stmt
{VARIABLESETSTMT
:kind 0
:name role
:args (
{A_CONST
:val "alice"
:location 9
}
)
:jumble_args false
:is_local false
:location 9
}
:stmt_location 0
:stmt_len 14
}

```

WARNING: ===== ANALYZED QUERY =====

```
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VARIABLESETSTMT
:kind 0
:name role
:args (
{A_CONST
:val "alice"
:location 9
}
)
:jumble_args false
:is_local false
:location 9
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 14
}
```

SET

WARNING: ===== RAW PARSE TREE =====

```
DETAIL: {RAWSTMT
:stmt
{EXPLAINSTMT
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTARTGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields (
{A_STAR
}
)
:location 43
}
}
:location 43
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 50
}
)
:whereClause <>
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:options (
{DEFELEM
:defnamespace <>
:defname analyze
:arg <>
:defaction 0
:location 9
}
{DEFELEM
:defnamespace <>
:defname verbose
:arg <>
:defaction 0
:location 18
}
{DEFELEM
:defnamespace <>
:defname buffers
:arg <>
:defaction 0
:location 27
}
)
}
:stmt_location 0
:stmt_len 56
}
```

WARNING: ===== ANALYZED QUERY =====

```
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
```

```

:utilityStmt
{EXPLAINSTMT
:query
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable (
      {RANGETBLENTTRY
        :alias <>
        :eref
          {ALIAS
            :aliasname orders
            :colnames ("o_orderkey" "o_custkey" "amount")
          }
        :rtekind 0
        :relid 16873
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCI true
        :securityQuals <>
      }
    )
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 16873
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 9 10)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
    :jointree
    {FROMEXPR
      :fromlist (
        {RANGETBLREF
          :rtindex 1
        }
      )
      :quals <>
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
      {TARGETENTRY
        :expr
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location 43
          }
        :resno 1
        :resname o_orderkey
        :ressortgroupref 0
        :resorigtbl 16873
        :resorigcol 1
        :resjunk false
      }
      {TARGETENTRY
        :expr
          {VAR
            :varno 1
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 2
            :location 43
          }
        :resno 2
        :resname o_custkey
        :ressortgroupref 0
        :resorigtbl 16873
        :resorigcol 2
        :resjunk false
      }
      {TARGETENTRY
        :expr
          {VAR
            :varno 1
            :varattno 3
            :vartype 1700
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 3
            :location 43
          }
        :resno 3
        :resname amount
        :ressortgroupref 0
        :resorigtbl 16873
        :resorigcol 3
        :resjunk false
      }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
  }
}

```

```

:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:options (
{DEFELEM
:defnamespace <>
:define analyze
:arg <>
:defaultion 0
:location 9
}
{DEFELEM
:defnamespace <>
:define verbose
:arg <>
:defaultion 0
:location 18
}
{DEFELEM
:defnamespace <>
:define buffers
:arg <>
:defaultion 0
:location 27
}
)
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 56
}

```

WARNING: ===== REWRITE INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "amount")
}
:rtekind 0
:reliid 16873
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reliid 16873
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
}
}
}

```

```

:varattnosyn 1
:location 43
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 43
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 43
}
:resno 3
:resname amount
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 3
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Table OID=16873

WARNING: ===== RLS POLICY ADDED =====

```

DETAIL: (
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_acctbal")
}
:rtekind 0
:relid 16865
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperinfos (
{RTEPERMISSIONINFO
:relid 16865
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals

```

```

{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)

```

WARNING: ===== REWRITE OUTPUT =====

```

DETAIL: (
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
}
)

```

```

:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders
        :colnames ("o_orderkey" "o_custkey" "amount")
      }
    :rtekind 0
    :relid 16873
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {SUBLINK
        :sublinkType 0
        :subLinkId 0
        :testexpr <>
        :opernName <>
        :subselect
          {QUERY
            :commandType 1
            :querySource 0
            :canSetTag true
            :utilityStmt <>
            :resultRelation 0
            :hasAggs false
            :hasWindowFuncs false
            :hasTargetSRFs false
            :hasSublinks false
            :hasDistinctOn false
            :hasRecursive false
            :hasModifyingCTE false
            :hasForUpdate false
            :hasRowSecurity false
            :hasGroupRTE false
            :isReturn false
            :cteList <>
            :rtable (
              {RANGETBLENTTRY
                :alias
                  {ALIAS
                    :aliasname c
                    :colnames <>
                  }
                :eref
                  {ALIAS
                    :aliasname c
                    :colnames ("c_custkey" "c_acctbal")
                  }
                :rtekind 0
                :relid 16865
                :inh true
                :relkind r
                :rellockmode 1
                :perminfoindex 1
                :tablesample <>
                :lateral false
                :inFromCl true
                :securityQuals <>
              }
            )
            :rteperminfos (
              {RTEPERMISSIONINFO
                :relid 16865
                :inh true
                :requiredPerms 2
                :checkAsUser 0
                :selectedCols (b 8 9)
                :insertedCols (b)
                :updatedCols (b)
              }
            )
          }
        :jointree
          {FROMEXPR
            :fromlist (
              {RANGETBLREF
                :rtindex 1
              }
            )
            :quals
              {BOOLEXP
                :boolop and
                :args (
                  {OPEXP
                    :opno 96
                    :opfuncid 65
                    :opresulttype 16
                    :opretset false
                    :opcollid 0
                    :inputcollid 0
                    :args (
                      {VAR
                        :varno 1
                        :varattno 1
                        :vartype 23
                        :vartypmod -1
                        :varcollid 0
                        :varnullingrels (b)
                        :varlevelsup 0
                        :varreturningtype 0
                        :varnosyn 1
                        :varrattnosyn 1
                        :location -1
                      }
                      {VAR
                        :varno 1
                        :varattno 2
                        :vartype 23
                        :vartypmod -1
                        :varcollid 0
                        :varnullingrels (b)
                        :varlevelsup 1
                        :varreturningtype 0
                        :varnosyn 1
                        :varrattnosyn 2
                        :location -1
                      }
                    )
                  }
                :location -1
              }
            )
            :quals
              {OPEXP
                :opno 1756
                :opfuncid 1720
                :opresulttype 16
                :opretset false
                :opcollid 0
                :inputcollid 0
                :args (
                  {VAR
                    :varno 1
                    :varattno 2

```

```

        :vartype 1700
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location -1
    }
    {FUNCEXP
    :funcid 1740
    :funcresulttype 1700
    :funcresultset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
        }
    )
    :location -1
}
}
:location -1
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :resortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
}
}
}
)
:rteperminfos (
    {RTEPERMISSIONINFO
    :relid 16873
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
    }
)
)
:jointree
{FROMEXPR
:fromlist (
    {RANGETBLREF
    :rtindex 1
    }
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 43
    }
    :resno 1
    :resname o_orderkey
    :resortgroupref 0
    :resorigtbl 16873
    :resorigcol 1
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    }
    }
}

```

```

:varattnosyn 2
:location 43
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 43
}
:resno 3
:resname amount
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 3
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
)

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "amount")
}
:rtekind 0
:relid 16873
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{SUBLINK
:sublinkType 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_acctbal")
}
:rtekind 0
:relid 16865
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16865
:inh true
:requiredPerms 2
:checkAsUser 0
}
)
)
}

```

```

:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromList (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1

```

```

        }
        :location -1
    }
}
)
: RTEPERMINFO (
{RTEPERMISSIONINFO
:reliid 16873
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 43
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 43
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 43
}
:resno 3
:resname amount
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 3
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: ===== FINAL PLAN =====
DETAIL: {SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 89
:scan.plan.plan_rows 25
:scan.plan.plan_width 13
:scan.plan.parallel_aware false
:scan.plan.parallel_safe false
:scan.plan.async_capable false
:scan.plan.plan_node_id 0
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 43
}
:resno 1

```

```

:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 43
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 43
}
:resno 3
:resname amount
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 3
:resjunk false
}
}
:scan.plan.qual (
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramIds (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 1.67
:per_call_cost 0.0025
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 1
}

```

QUERY PLAN

```

Seq Scan on public.orders (cost=0.00..89.00 rows=25 width=13) (actual time=0.059..0.073 rows=19.00 loops=1)
  Output: orders.o_orderkey, orders.o_custkey, orders.amount
  Filter: (ANY (orders.o_custkey = (hashed SubPlan 2).col1))
  Rows Removed by Filter: 31
  Buffers: shared hit=2
  SubPlan 2
    -> Seq Scan on public.customer c (cost=0.00..1.62 rows=18 width=4) (actual time=0.015..0.027 rows=18.00 loops=1)
      Output: c.c_custkey
      Filter: (c.c_acctbal > '5000'::numeric)
      Rows Removed by Filter: 32
      Buffers: shared hit=1

```

```

Planning:
  Buffers: shared hit=79
Planning Time: 2.219 ms
Execution Time: 0.138 ms
(15 rows)

```

```
udf_testing=> RESET ROLE;
```

```
DROP POLICY IF EXISTS orders_rls_policy ON orders;
```

```
CREATE OR REPLACE FUNCTION customer_acctbal_check(custkey INT)
RETURNS BOOLEAN
LANGUAGE SQL
STABLE
AS $$

```

```

  SELECT EXISTS (
    SELECT 1
    FROM customer c
    WHERE c.c_custkey = custkey
    AND c.c_acctbal > 5000
  );

```

```
$$;
```

```
CREATE POLICY orders_rls_policy
```

```

ON orders
USING (
    customer_acctbal_check(o_custkey)
);

SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT *
FROM orders;
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
  {VARIABLESETSTMT
    :kind 4
    :name role
    :args <>
    :jumble_args false
    :is_local false
    :location -1
  }
:stmt_location 0
:stmt_len 10
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
  {VARIABLESETSTMT
    :kind 4
    :name role
    :args <>
    :jumble_args false
    :is_local false
    :location -1
  }
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 10
}

RESET
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
  {DROPTSTMT
    :objects (("orders" "orders_rls_policy"))
    :removeType 28
    :behavior 0
    :missing_ok true
    :concurrent false
  }
:stmt_location 0
:stmt_len 49
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
  {DROPTSTMT
    :objects (("orders" "orders_rls_policy"))
    :removeType 28
    :behavior 0
    :missing_ok true
    :concurrent false
  }
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>

```

```

:withCheckOptions <>
:stmt_location 0
:stmt_len 49
}

DROP POLICY
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{CREATEFUNCTIONSTMT
:is_procedure false
:replace true
:funcname ("customer_acctbal_check")
:parameters {
{FUNCTIONPARAMETER
:name custkey
:argType
{TYPPENAME
:names ("pg_catalog" "int4")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 58
}
:mode 100
:defexpr <>
:location 50
}
}
:returnType
{TYPPENAME
:names ("pg_catalog" "bool")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 71
}
:options {
{DEFELEM
:defnamespace <>
:defname language
:arg "sql"
:defaction 0
:location 79
}
{DEFELEM
:defnamespace <>
:defname volatility
:arg "stable"
:defaction 0
:location 92
}
{DEFELEM
:defnamespace <>
:defname as
:arg ("
SELECT EXISTS\ \
\ \ \ \ \ \ \ \ SELECT\ 1\
\ \ \
\ \ \ \ \ \ \ \ FROM\ customer\ c\
\ \ \ \ \ \ \ \ WHERE\ c.c_custkey\ =\
custkey\
\ \ \ \ \ \ \ \ AND\ c.c_acctbal\ >\ 5000\
\ \ \ \ \ \ \ \ )\
;\
")
:defaction 0
:location 99
}
}
:sql_body <>
}
:stmt_location 0
:stmt_len 244
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{CREATEFUNCTIONSTMT
:is_procedure false
:replace true
:funcname ("customer_acctbal_check")
:parameters {
{FUNCTIONPARAMETER
:name custkey
:argType
{TYPPENAME
:names ("pg_catalog" "int4")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 58
}
:mode 100
:defexpr <>
:location 50
}
}
:returnType
{TYPPENAME
:names ("pg_catalog" "bool")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 71
}
:options {
{DEFELEM
:defnamespace <>
:defname language
:arg "sql"
:defaction 0
:location 79
}
{DEFELEM
:defnamespace <>
:defname volatility
:arg "stable"
:defaction 0
:location 92
}
{DEFELEM
:defnamespace <>
:defname as
:arg ("
SELECT EXISTS\ \
\ \ \ \ \ \ \ \ SELECT\ 1\
\ \ \
\ \ \ \ \ \ \ \ FROM\ customer\ c\
\ \ \ \ \ \ \ \ WHERE\ c.c_custkey\ =\
custkey\
\ \ \ \ \ \ \ \ AND\ c.c_acctbal\ >\ 5000\

```

```

\ \ \ \)
; \
")
    :defaction 0
    :location 99
    }
    )
    :sql_body <>
    }
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :ctelist <>
    :rtable <>
    :rteperminfos <>
    :jointree <>
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList <>
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 244
    }

```

WARNING: ===== REWRITE INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree
{FROMEXPR
:fromlist <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_acctbal")
}
:rtekind 0
:relid 16865
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16865
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1

```

```

    }
  )
  :quals
  {BOOLEXP
  :boolop and
  :args (
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
  {VAR
  :varno 1
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 1
  :location 76
  }
  {PARAM
  :paramkind 0
  :paramid 1
  :paramtype 23
  :paramtypmod -1
  :paramcollid 0
  :location 90
  }
  )
  :location 88
  }
  {OPEXP
  :opno 1756
  :opfuncid 1720
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
  {VAR
  :varno 1
  :varattno 2
  :vartype 1700
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 2
  :location 112
  }
  {FUNCEXP
  :funcid 1740
  :funcresulttype 1700
  :funcresultset false
  :funcvariadic false
  :funcformat 2
  :funccollid 0
  :inputcollid 0
  :args (
  {CONST
  :consttype 23
  :constypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull false
  :location 126
  :constvalue 4 [ -128 19 0 0 0 0 0 0 ]
  }
  )
  :location -1
  }
  )
  :location 124
  )
  :location 108
  )
  :mergeActionList <>
  :mergeTargetRelation 0
  :mergeJoinCondition <>
  :targetList (
  {TARGETENTRY
  :expr
  {CONST
  :consttype 23
  :constypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull false
  :location 36
  :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
  }
  :resno 1
  :resname ?column?
  :resortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
  )
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDeps <>
  :withCheckOptions <>
  :stmt_location 0
  :stmt_len 0
  }
  :location 12
  )
  :resno 1
  :resname exists
  :resortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
  )
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>

```

```

:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}

WARNING: ===== REWRITE OUTPUT =====
DETAIL: (
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree
{FROMEXPR
:fromList <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_acctbal")
}
:rtekind 0
:relid 16865
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16865
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromList (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 76
}
{PARAM
:paramkind 0
:paramid 1
:paramtype 23
:paramtypmod -1

```

```

        :paramcollid 0
        :location 90
    }
}
:location 88
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varsetno 2
:varstype 1700
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 112
}
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 126
:constvalue 4 [ -128 19 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location 124
}
)
:location 108
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 36
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 12
}
:resno 1
:resname exists
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}
)

```

```

CREATE FUNCTION
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{CREATEPOLICYSTMT
:policy_name orders_rls_policy
:table
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true

```

```

:relpersistence p
:alias <>
:location 35
}
:cmd_name all
:permissive true
:roles (
  {ROLESPEC
    :roletype 4
    :rolename <>
    :location -1
  }
)
:qual
{FUNCCALL
  :funcname ("customer_acctbal_check")
  :args (
    {COLUMNREF
      :fields ("o_custkey")
      :location 77
    }
  )
  :agg_order <>
  :agg_filter <>
  :over <>
  :agg_within_group false
  :agg_star false
  :agg_distinct false
  :func_variadic false
  :funcformat 0
  :location 54
}
:with_check <>
}
:stmt_location 0
:stmt_len 89
}

```

WARNING: ===== ANALYZED QUERY =====

```

DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{CREATEPOLICYSTMT
:policy_name orders_rls_policy
:table
{RANGEVAR
:catalogname <>
:schemaname <>
:rolname orders
:inh true
:relpersistence p
:alias <>
:location 35
}
:cmd_name all
:permissive true
:roles (
  {ROLESPEC
    :roletype 4
    :rolename <>
    :location -1
  }
)
:qual
{FUNCCALL
  :funcname ("customer_acctbal_check")
  :args (
    {COLUMNREF
      :fields ("o_custkey")
      :location 77
    }
  )
  :agg_order <>
  :agg_filter <>
  :over <>
  :agg_within_group false
  :agg_star false
  :agg_distinct false
  :func_variadic false
  :funcformat 0
  :location 54
}
:with_check <>
}
:resultRelation 0
:hasAggs false
:hasWindowfuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 89
}

```

CREATE POLICY

WARNING: ===== RAW PARSE TREE =====

```

DETAIL: {RAWSTMT
:stmt
{VARIABLESETSTMT
:kind 0
:name role
:args (
  {A_CONST
    :val "alice"
    :location 9
  }
)
:jumble_args false
:is_local false
:location 9
}
:stmt_location 0
:stmt_len 14
}

```

```

}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VARIABLESETSTMT
:kind 0
:name role
:args (
{A_CONST
:val "alice"
:location 9
}
)
:jumble.args false
:is_local false
:location 9
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rtepermInfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 14
}

SET
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{EXPLAINSTMT
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RETARGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields (
{A_STAR
}
)
:location 43
}
:location 43
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 50
}
)
:whereClause <>
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:options (
{DEFELEM
:defnamespace <>
:defname analyze
:arg <>
:defaultion 0
:location 9
}
{DEFELEM
:defnamespace <>
:defname verbose
:arg <>
:defaultion 0
:location 18
}
{DEFELEM
:defnamespace <>
:defname buffers
:arg <>
:defaultion 0
:location 27
}
)
:stmt_location 0
:stmt_len 56
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0

```

```

:canSetTag true
:utilityStmt
{EXPLAIN_STMT
  :query
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable (
      {RANGE_TBL_ENTRY
        :alias <>
        :eref
        {ALIAS
          :aliasname orders
          :colnames ("o_orderkey" "o_custkey" "amount")
        }
        :rtekind 0
        :relid 16873
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCI true
        :securityQuals <>
      }
    )
    :rttperminfos (
      {RTE_PERMISSION_INFO
        :relid 16873
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 9 10)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
    :jointree
    {FROMEXPR
      :fromlist (
        {RANGE_TBL_REF
          :rtindex 1
        }
      )
      :quals <>
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
      {TARGET_ENTRY
        :expr
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 43
        }
        :resno 1
        :resname o_orderkey
        :ressortgroupref 0
        :resorigtbl 16873
        :resorigcol 1
        :resjunk false
      }
      {TARGET_ENTRY
        :expr
        {VAR
          :varno 1
          :varattno 2
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 2
          :location 43
        }
        :resno 2
        :resname o_custkey
        :ressortgroupref 0
        :resorigtbl 16873
        :resorigcol 2
        :resjunk false
      }
      {TARGET_ENTRY
        :expr
        {VAR
          :varno 1
          :varattno 3
          :vartype 1700
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 3
          :location 43
        }
        :resno 3
        :resname amount
        :ressortgroupref 0
        :resorigtbl 16873
        :resorigcol 3
        :resjunk false
      }
    )
  }
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0

```

```

:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:options (
{DEFELEM
:defnamespace <>
:dbname analyze
:arg <>
:defaction 0
:location 9
}
{DEFELEM
:defnamespace <>
:dbname verbose
:arg <>
:defaction 0
:location 18
}
{DEFELEM
:defnamespace <>
:dbname buffers
:arg <>
:defaction 0
:location 27
}
)
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 56
}

```

WARNING: ===== REWRITE INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "amount")
}
:rtekind 0
:relid 16873
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16873
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
}
}
)
}

```

```

:varnosyn 1
:varattnosyn 1
:location 43
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varstype 23
:varstype -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 43
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varstype 1700
:varstype -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 43
}
:resno 3
:resname amount
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 3
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: Table OID=16873
WARNING: ===== RLS POLICY ADDED =====
DETAIL: {
{FUNCEXP
:funcid 16887
:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 2
:varstype 23
:varstype -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
}

WARNING: ===== REWRITE OUTPUT =====
DETAIL: {
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:table {
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "amount")
}
:rtekind 0
:relid 16873
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{FUNCEXP

```

```

:funcid 16887
:funcresulttype 16
:funcreset false
:funcvariadic false
:funcformat 0
:vartypmod -1
:varcollid 0
:inputcollid 0
:args (
  (VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  )
)
:location -1
)
)
):rtperminfos (
  (RTPERMISSIONINFO
    :relid 16873
    :inh true
    :requiredPerms 2
    :FROMEXPR
    :fromlist (
      (RANGETBLREF
        :rtindex 1
      )
    )
    :quals <>
  )
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  (TARGETENTRY
    :expr
    (VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 43
    )
    :resno 1
    :resname o_orderkey
    :ressortgroupref 0
    :resorigtbl 16873
    :resorigcol 1
    :resjunk false
  )
  (TARGETENTRY
    :expr
    (VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 43
    )
    :resno 2
    :resname o_custkey
    :ressortgroupref 0
    :resorigtbl 16873
    :resorigcol 2
    :resjunk false
  )
  (TARGETENTRY
    :expr
    (VAR
      :varno 1
      :varattno 3
      :vartype 1700
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location 43
    )
    :resno 3
    :resname amount
    :ressortgroupref 0
    :resorigtbl 16873
    :resorigcol 3
    :resjunk false
  )
)
):override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmtLength 0
:stmt_len 0
)
)
)
WARNING: ===== PLANNER INPUT =====
DETAIL:
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>

```

```

:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders
        :colnames ("o_orderkey" "o_custkey" "amount")
      }
    :rtekind 0
    :reldid 16873
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals (
      {FUNCEXP
        :funcid 16887
        :funcresulttype 16
        :funcresultset false
        :funcvariadic false
        :funcformat 0
        :funccollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 2
            :location -1
          }
        )
        :location -1
      }
    )
  }
)
:rtepermissions (
  {RTEPERMISSIONINFO
    :reldid 16873
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals <>
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 43
      }
    :resno 1
    :resname o_orderkey
    :ressortgroupref 0
    :resorigtbl 16873
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location 43
      }
    :resno 2
    :resname o_custkey
    :ressortgroupref 0
    :resorigtbl 16873
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 3
        :vartype 1700
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location 43
      }
    :resno 3
    :resname amount
    :ressortgroupref 0
    :resorigtbl 16873
    :resorigcol 3
    :resjunk false
  }
)

```

```

    }
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMasks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: ===== FINAL PLAN =====
DETAIL: {SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 14
:scan.plan.plan_rows 17
:scan.plan.plan_width 13
:scan.plan.parallel_aware false
:scan.plan.parallel_safe false
:scan.plan.async_capable false
:scan.plan.plan_node_id 0
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 43
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 43
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 43
}
:resno 3
:resname amount
:ressortgroupref 0
:resorigtbl 16873
:resorigcol 3
:resjunk false
}
}
:scan.plan.qual (
{FUNCEXP
:funcid 16887
:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 1
}

WARNING: ===== REWRITE INPUT =====
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false

```

```

:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree
{FROMEXPR
:fromlist <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_acctbal")
}
:rtekind 0
:relid 16865
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16865
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 76
}
)
{PARAM
:paramkind 0
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location 90
}
)
:location 88
}
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:varatype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 112
}
)
{FUNCExpr
:funcid 1740

```

```

:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 126
    :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
  }
)
:location -1
}
)
:location 124
}
)
:location 108
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location 36
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :resortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 12
}
:resno 1
:resname exists
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}

```

WARNING: ===== REWRITE OUTPUT =====

```

DETAIL: (
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks true
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable <>
    :rteperminfos <>
    :jointree
    {FROMEXPR
      :fromlist <>
      :quals <>
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
      {TARGETENTRY
        :expr
        {SUBLINK
          :sublinkType 0
          :sublinkId 0
          :testexpr <>
          :operName <>
          :subselect
          {QUERY
            :commandType 1
            :querySource 0
          }
        }
      }
    )
  }
)

```

```

:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname c
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname c
        :colnames ("c_custkey" "c_acctbal")
      }
    :rtekind 0
    :relid 16865
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16865
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 76
      }
      {PARAM
        :paramkind 0
        :paramid 1
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location 90
      }
    )
    :location 88
  }
  {OEXPR
    :opno 1756
    :opfuncid 1720
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 2
        :vartype 1700
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location 112
      }
      {FUNCEXP
        :funcid 1740
        :funcresulttype 1700
        :funcset false
        :funcvariadic false
        :funcformat 2
        :funccollid 0
        :inputcollid 0
        :args (
          {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location 120
            :constvalue 4 [ -120 19 0 0 0 0 0 ]
          }
        )
        :location -1
      }
    )
    :location 124
  }
)
:location 108
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY

```

```

        :expr
        {CONST
         :consttype 23
         :consttypmod -1
         :constcollid 0
         :constlen 4
         :constbyval true
         :constisnull false
         :location 36
         :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :resortgrouppref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 12
}
:resno 1
:resname exists
:resortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}
)

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rtepermInfos <>
:jointree
{FROMEXPR
:fromlist <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGEBTLENTY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_acctbal")
}
:rtekind 0
:relid 16865
:inh true
:relkind r
:rellockmode 1

```

```

:perminfoindex 1
:tablesampl <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16865
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 76
}
)
{PARAM
:paramkind 0
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location 90
}
)
:location 88
}
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 1700
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 112
}
)
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 126
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location 124
}
)
:location 108
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 36
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>

```

```

:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 12
}
:resno 1
:resname exists
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}

WARNING: ===== FINAL PLAN =====
DETAIL: {RESULT
:plan.disabled_nodes 0
:plan.startup_cost 1.75
:plan.total_cost 1.76
:plan.plan_rows 1
:plan.plan_width 1
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
:resno 1
:resname exists
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree <>
:plan.righttree <>
:plan.initPlan (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name InitPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe true
:setParam (1 0)
:parParam <>
:args <>
:startup_cost 0
:per_call_cost 1.75
}
)
:plan.extParam (b)
:plan.allParam (b 0)
:resconstantqual <>
}

```

```

WARNING: ===== PLANNER INPUT =====
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree
{FROMEXPR
:fromlist <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
}
}
}
}

```

```

:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname c
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname c
      :colnames ("c_custkey" "c_acctbal")
    }
    :rtekind 0
    :relid 16865
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16865
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
  :quals
  {BOOLEXP
    :boolop and
    :args (
      {OEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location 76
          }
          {PARAM
            :paramkind 0
            :paramid 1
            :paramtype 23
            :paramtypmod -1
            :paramcollid 0
            :location 90
          }
        )
        :location 88
      }
      {OEXPR
        :opno 1756
        :opfuncid 1720
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 2
            :vartype 1700
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 2
            :location 112
          }
          {FUNCEXP
            :funcid 1740
            :funcresulttype 1700
            :funcset false
            :funcvariadic false
            :funcformat 2
            :funccollid 0
            :inputcollid 0
            :args (
              {CONST
                :consttype 23
                :consttypmod -1
                :constcollid 0
                :constlen 4
                :constbyval true
                :constisnull false
                :location 126
                :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
              }
            )
            :location -1
          }
        )
        :location 124
      }
    )
    :location 108
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
    }
  }
)

```

```

        :constisnull false
        :location 36
        :constvalue 4 [ 1 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 12
}
:resno 1
:resname exists
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}
)
WARNING: ===== FINAL PLAN =====
DETAIL: {RESULT
:plan.disabled_nodes 0
:plan.startup_cost 1.75
:plan.total_cost 1.76
:plan.plan_rows 1
:plan.plan_width 1
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
:resno 1
:resname exists
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree <>
:plan.righttree <>
:plan.initPlan (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name InitPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe true
:setParam (i 0)
:parParam <>
:args <>
:startup_cost 0
:per_call_cost 1.75
}
)
:plan.extParam (b)
:plan.allParam (b 0)
:resconstantqual <>
}

```

```

WARNING: ===== PLANNER INPUT =====
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree
{FROMEXPR

```

```

:fromlist <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
  :expr
  {SUBLINK
  :sublinkType 0
  :sublinkId 0
  :testexpr <>
  :operName <>
  :subselect
  {QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :cteList <>
  :rtable (
    {RANGETBLENTY
    :alias
    {ALIAS
    :aliasname c
    :colnames <>
    }
    :eref
    {ALIAS
    :aliasname c
    :colnames ("c_custkey" "c_acctbal")
    }
    :rtekind 0
    :relid 16865
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
  )
  :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 16865
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
    }
  )
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 76
    }
    {PARAM
    :paramkind 0
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location 90
    }
  )
  :location 88
  }
  {OPEXP
  :opno 1756
  :opfuncid 1720
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 2
    :vartype 1700
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 112
    }
    {FUNCEXP
    :funcid 1740
    :funcresulttype 1700
    :funcset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false

```

```

        :location 126
        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
    }
}
:location -1
}
)
:location 124
}
)
:location 108
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 36
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 12
}
:resno 1
:resname exists
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}

WARNING: ===== FINAL PLAN =====
DETAIL: {RESULT
:plan.disabled_nodes 0
:plan.startup_cost 1.75
:plan.total_cost 1.76
:plan.plan_rows 1
:plan.plan_width 1
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
    {TARGETENTRY
    :expr
    {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 16
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
    :resno 1
    :resname exists
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
)
:plan.qual <>
:plan.lefttree <>
:plan.righttree <>
:plan.initPlan (
    {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramIds <>
    :plan_id 1
    :plan_name InitPlan\ 1
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEq false
    :parallel_safe true
    :setParameter (i 0)
    :parParam <>
    :args <>
    :startup_cost 0
    :per_call_cost 1.75
    }
)
:plan.extParam (b)
:plan.allParam (b 0)
:resconstantqual <>

```

```

}

WARNING: ===== PLANNER INPUT =====
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree
{FROMEXPR
:fromlist <>
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_acctbal")
}
:rtekind 0
:relid 16865
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16865
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 76
}
{PARAM
:paramkind 0
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location 90
}
)
:location 88
}
{OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:varatype 1700

```

```

        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location 112
    }
    {FUNCEXP
    :funcid 1740
    :funcresulttype 1700
    :funcresultset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 126
        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
        }
    )
    :location -1
    }
    )
    :location 124
    }
    )
    :location 108
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 36
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
    }
    :location 12
    }
    :resno 1
    :resname exists
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 5
    :stmt_len 131
    }
}

WARNING: ===== FINAL PLAN =====
DETAIL: {RESULT
:plan.disabled_nodes 0
:plan.startup_cost 1.75
:plan.total_cost 1.76
:plan.plan_rows 1
:plan.plan_width 1
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
:resno 1
:resname exists
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>

```

```

:plan.lefttree <>
:plan.righttree <>
:plan.initPlan (
  {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramIds <>
    :plan_id 1
    :plan_name InitPlan\ 1
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse false
    :parallel_safe true
    :setParam (1 0)
    :parParam <>
    :args <>
    :startup_cost 0
    :per_call_cost 1.75
  }
)
:plan.extParam (b)
:plan.allParam (b 0)
:resconstantqual <>
}

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree
  {FROMEXPR
    :fromlist <>
    :quals <>
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
    :expr
      {SUBLINK
        :subLinkType 0
        :subLinkId 0
        :testexpr <>
        :opName <>
        :subselect
          {QUERY
            :commandType 1
            :querySource 0
            :canSetTag true
            :utilityStmt <>
            :resultRelation 0
            :hasAggs false
            :hasWindowFuncs false
            :hasTargetSRFs false
            :hasSubLinks false
            :hasDistinctOn false
            :hasRecursive false
            :hasModifyingCTE false
            :hasForUpdate false
            :hasRowSecurity false
            :hasGroupRTE false
            :isReturn false
            :cteList <>
            :rtable (
              {RANGETBLENTY
                :alias
                  {ALIAS
                    :aliasname c
                    :colnames <>
                  }
                :eref
                  {ALIAS
                    :aliasname c
                    :colnames ("c_custkey" "c_acctbal")
                  }
                :rtekind 0
                :relid 16865
                :inh true
                :relkind r
                :rellockmode 1
                :perminfoindex 1
                :tablesample <>
                :lateral false
                :inFromCl true
                :securityQuals <>
              }
            )
          }
        :rteperminfos (
          {RTEPERMISSIONINFO
            :relid 16865
            :inh true
            :requiredPerms 2
            :checkAsUser 0
            :selectedCols (b 8 9)
            :insertedCols (b)
            :updatedCols (b)
          }
        )
      }
    :jointree
      {FROMEXPR
        :fromlist (
          {RANGETBLREF
            :rtindex 1
          }
        )
      }
    :quals
      {BOOLEXP
        :boolop and
        :args (
          {OPEXP
            :opno 96
            :opfuncid 65
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 1
              }
            )
          }
        )
      }
    }
  }
}

```

```

        :location 76
      }
      {PARAM
      :paramkind 0
      :paramid 1
      :paramtype 23
      :paramtypmod -1
      :paramcollid 0
      :location 90
      }
    }
    :location 88
  }
  {OPEXPR
  :opno 1756
  :opfuncid 1720
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
  {VAR
  :varno 1
  :varattno 2
  :vartype 1700
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 2
  :location 112
  }
  {FUNCEXPR
  :funcid 1740
  :funcresulttype 1700
  :funcsetset false
  :funcvariadic false
  :funcformat 2
  :funccollid 0
  :inputcollid 0
  :args (
  {CONST
  :consttype 23
  :consttypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull false
  :location 126
  :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
  }
  )
  :location -1
  }
  :location 124
}
)
:location 108
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 36
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 12
}
:resno 1
:resname exists
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}
)

```

WARNING: ===== FINAL PLAN =====

DETAIL: {RESULT
:plan.disabled_nodes 0
:plan.startup_cost 1.75
:plan.total_cost 1.76
:plan.plan_rows 1

```

:plan.plan_width 1
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
  {TARGETENTRY
    :expr
    {PARAM
      :paramkind 1
      :paramid 0
      :paramtype 16
      :paramtypmod -1
      :paramcollid 0
      :location -1
    }
    :resno 1
    :resname exists
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree <>
:plan.righttree <>
:plan.initPlan (
  {SUBPLAN
    :sublinkType 0
    :testexpr <>
    :paramids <>
    :plan_id 1
    :plan_name InitPlan\ 1
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse false
    :parallel_safe true
    :setParam (i 0)
    :parParam <>
    :args <>
    :startup_cost 0
    :per_call_cost 1.75
  }
)
:plan.extParam (b)
:plan.allParam (b 0)
:resconstantqual <>
}

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree
  {FROMEXPR
    :fromlist <>
    :quals <>
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {SUBLINK
      :sublinkType 0
      :sublinkId 0
      :testexpr <>
      :operName <>
      :subselect
        {QUERY
          :commandType 1
          :querySource 0
          :canSetTag true
          :utilityStmt <>
          :resultRelation 0
          :hasAggs false
          :hasWindowFuncs false
          :hasTargetSRFs false
          :hasSublinks false
          :hasDistinctOn false
          :hasRecursive false
          :hasModifyingCTE false
          :hasForUpdate false
          :hasRowSecurity false
          :hasGroupRTE false
          :isReturn false
          :ctelist <>
          :rtable (
            {RANGETBENTRY
              :alias
              {ALIAS
                :aliasname c
                :colnames <>
              }
              :eref
              {ALIAS
                :aliasname c
                :colnames ("c_custkey" "c_acctbal")
              }
              :rtekind 0
              :relid 16865
              :inh true
              :relkind r
              :rellockmode 1
              :perminfoindex 1
              :tablesample <>
              :lateral false
              :inFromCi true
              :securityQuals <>
            }
          )
          :rteperminfos (
            {RTEPERMISSIONINFO
              :relid 16865
              :inh true
              :requiredPerms 2
              :checkAsUser 0
              :selectedCols (b 8 9)
              :insertedCols (b)
              :updatedCols (b)
            }
          )
        }
      }
    }
  }
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1

```

```

    }
  )
  :quals
  {BOOLEXP
  :boolop and
  :args (
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
  {VAR
  :varno 1
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 1
  :location 76
  }
  {PARAM
  :paramkind 0
  :paramid 1
  :paramtype 23
  :paramtypmod -1
  :paramcollid 0
  :location 90
  }
  )
  :location 88
  }
  {OPEXP
  :opno 1756
  :opfuncid 1720
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
  {VAR
  :varno 1
  :varattno 2
  :vartype 1700
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 2
  :location 112
  }
  {FUNCEXP
  :funcid 1740
  :funcresulttype 1700
  :funcresultset false
  :funcvariadic false
  :funcformat 2
  :funccollid 0
  :inputcollid 0
  :args (
  {CONST
  :consttype 23
  :constypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull false
  :location 126
  :constvalue 4 [ -128 19 0 0 0 0 0 0 ]
  }
  )
  :location -1
  }
  )
  :location 124
  )
  :location 108
  )
  :mergeActionList <>
  :mergeTargetRelation 0
  :mergeJoinCondition <>
  :targetList (
  {TARGETENTRY
  :expr
  {CONST
  :consttype 23
  :constypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull false
  :location 36
  :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
  }
  :resno 1
  :resname ?column?
  :resortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
  )
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDeps <>
  :withCheckOptions <>
  :stmt_location 0
  :stmt_len 0
  )
  :location 12
  )
  :resno 1
  :resname exists
  :resortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>

```

```
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 5
:stmt_len 131
}

WARNING: ===== FINAL PLAN =====
DETAIL: {RESULT
:plan.disabled_nodes 0
:plan.startup_cost 1.75
:plan.total_cost 1.76
:plan.plan_rows 1
:plan.plan_width 1
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist {
{TARGETENTRY
:expr
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
:resno 1
:resname exists
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree <>
:plan.righttree <>
:plan.initPlan (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 1
:plan_name InitPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe true
:setParam (i 0)
:parParam <>
:args <>
:startup_cost 0
:per_call_cost 1.75
}
)
:plan.extParam (b)
:plan.allParam (b 0)
:resconstantqual <>
}
```

QUERY PLAN

```
Seq Scan on public.orders (cost=0.00..14.00 rows=17 width=13) (actual time=7.925..10.127 rows=19.00 loops=1)
Output: o_orderkey, o_custkey, amount
Filter: customer_acctbal_check(orders.o_custkey)
Rows Removed by Filter: 31
Buffers: shared hit=51
Planning:
Buffers: shared hit=4
Planning Time: 0.396 ms
Execution Time: 10.154 ms
(9 rows)

udf_testing=>
```